

TensorFVM: analytic verification of incompressible finite-volume solvers

Abstract

Two independent solver adapters share one benchmark pipeline. Plane Poiseuille flow tests a no-slip steady channel; a periodic Taylor-Green vortex tests transient face momentum. Every declared regular run must have velocity, pressure and applicable wall errors strictly below 3%, together with separate solver/mass/energy gates. Negative controls remain archived. Overall regular acceptance: True.

1. Mathematical model and boundary conditions

Incompressible momentum: $\text{div}(\mathbf{u})=0$; $\rho(\frac{d\mathbf{u}}{dt}+\text{div}(\mathbf{u} \otimes \mathbf{u}))=-\text{grad}(p)+\text{div}(\mu(\text{grad}(\mathbf{u})+\text{grad}(\mathbf{u})^T))$. All quantities are SI.

Channel: $L=0.12$ m, $H=0.02$ m, $\rho=1000$ kg/m³, $\nu=1\text{e-}5$ m²/s, $U_b=5\text{e-}4$ m/s, $Re_H=1$. Uniform velocity inlet, no-slip walls and zero gauge-pressure outlet. Analytic comparison is confined to developed stations $3H \leq x \leq 5H$. Reference: $u=6U_b(y/H)(1-y/H)$; $dp/dx=-12\mu U_b/H^2$; $|\tau_w|=6\mu U_b/H$.

Periodic vortex: $[0, 2\pi]^3$ m; $\rho=1$ kg/m³, $\mu=0.1$ Pa s, $U_0=1$ m/s.
 $u=\sin(x)\cos(y)\exp(-2\nu t)$; $v=-\cos(x)\sin(y)\exp(-2\nu t)$; $w=0$.
 $p=\rho[\cos(2x)+\cos(2y)]\exp(-4\nu t)/4$. It is a 2D exact solution embedded in four z layers, not 3D turbulence. Pressure is compared at the last midpoint $t-\Delta t/2$, with zero mean.

2. Numerical methods and verification protocol

Channel: staggered finite volumes and SIMPLE, half-cell wall diffusion, tolerance $1e-6$.

Grids 36×12 , 72×24 and 144×48 . Vortex: authoritative MAC face velocities, shared centered dual-volume momentum fluxes, complete symmetric stress, compatible periodic FFT pressure projection and implicit midpoint. Spatial grids $32 \times 32 \times 4$, $48 \times 48 \times 4$, $64 \times 64 \times 4$ at $dt=0.0125$ s. Time steps 0.1, 0.05 and 0.025 s on $32 \times 32 \times 4$. All vortex runs end at 0.5 s.

Relative L2 uses global reference norm; relative Linf uses reference peak. Pressure gauge/time/locations are explicitly aligned. Temporal order uses a fixed-grid exact discrete eigenmode exponential; this is not a continuous physical reference. Observed order is computed from actual errors and refinement ratios. No assumed-order GCI is presented. Negative controls: unconverged channel and coarse vortex pressure failure.

3. Results and discussion

Regular-run acceptance: True.

Observed orders: {"channel_velocity": [1.969562587297749, 1.9184359357168919],
"channel_pressure_gradient": [1.9934624306292263, 2.134205788719911], "tg_velocity":
[1.9989589105828733, 2.000226557168189], "tg_pressure": [1.9968077082851627,
1.999166343016587], "tg_time": [2.0000627255879495, 2.000015610195456]}

The following table reports actual computed errors. All detailed numerical gates, configurations and clocks are in result.json. Fields.npz stores authoritative and visualization fields; full accepted vortex face states and iteration histories are retained. Figures use actual computed values, physical coordinates and labelled SI color bars.

Case	Study	Grid	dt / Ma	u L2 (%)	p (%)	Gate
poiseuille	spatial	36x12	--	0.63436	1.36985	PASS
poiseuille	spatial	72x24	--	0.16197	0.34402	PASS
poiseuille	spatial	144x48	--	0.04285	0.07836	PASS
taylor-green	spatial	32x32x4	dt=0.0125	0.03209	0.89781	PASS
taylor-green	spatial	48x48x4	dt=0.0125	0.01427	0.39954	PASS
taylor-green	spatial	64x64x4	dt=0.0125	0.00802	0.22480	PASS
taylor-green	temporal	32x32x4	dt=0.1	0.03176	0.89426	PASS
taylor-green	temporal	32x32x4	dt=0.05	0.03201	0.89803	PASS
taylor-green	temporal	32x32x4	dt=0.025	0.03207	0.89817	PASS
poiseuille	negative-control	36x12	--	25.35620	74.13972	FAIL
taylor-green	negative-control	16x16x4	dt=0.0125	0.12793	3.56268	FAIL
taylor-green-lbm	comparison	32x32	Ma=0.050	0.63348	0.86173	PASS
taylor-green-lbm	comparison	64x64	Ma=0.050	0.15207	0.81428	PASS
taylor-green-lbm	comparison	32x32	Ma=0.025	0.64003	1.68311	PASS

4. Scope, reproducibility and conclusions

The 3% gate is a declared project analytic-verification target, not a universal publication or safety standard. Channel entrance flow is excluded from developed-flow comparisons. Vortex periodic accuracy does not establish solid-wall, free-surface, moving-body, high-Re turbulence, airfoil or SUBOFF accuracy. Repository tensorlbm experiments are not used as these solver backends.

Install: `python -m pip install -e '[benchmark,dev]'`

Run: `python -m tensorfvm.verification --output docs/verification-benchmarks`

Audit: `python -m tensorfvm.verification.audit --directory docs/verification-benchmarks`

Source hashes, git base, environment and artifact hashes are supplied. PNG at 300 dpi and PDF/SVG figures, CSV curves, raw NPZ, full JSON histories and editable TeX accompany this manuscript. The analytic references are derived directly from the governing equations; discrete time references use the compatible Fourier eigenvalue.

A separate 128x128x4 run with $dt=0.0125$ was rejected by the Picard nonlinear solver at its third step, after 0.025 s. Nonfinite candidate fields and unchanged accepted state are retained. The qualified 32/48/64 study does not certify that 128 grid.

References: Taylor and Green, Proc. R. Soc. A 158 (1937), 499-521, DOI 10.1098/rspa.1937.0036 (classical vortex-verification background; this report uses a 2D decay variant). NASA NPARC, Examining Spatial (Grid) Convergence, grc.nasa.gov/www/wind/valid/tutorial/spatconv.html (methodology, not the source of a 3% standard).

This verification report requires author details, journal formatting and peer review before submission.

5. Matched TensorLBM comparison

Same continuum periodic domain, rho, viscosity, velocity amplitude and end time. Analytic errors use each solver's actual DOF locations and correct pressure clock. LBM uses external production D2Q9 equilibrium/BGK/stream, without mass correction. Float64 populations retain the existing upstream float32 weight constants.

LBM 32x32, Ma=0.0496: u L2=0.63348%, p L2=0.86173%, passed=True

LBM 64x64, Ma=0.0498: u L2=0.15207%, p L2=0.81428%, passed=True

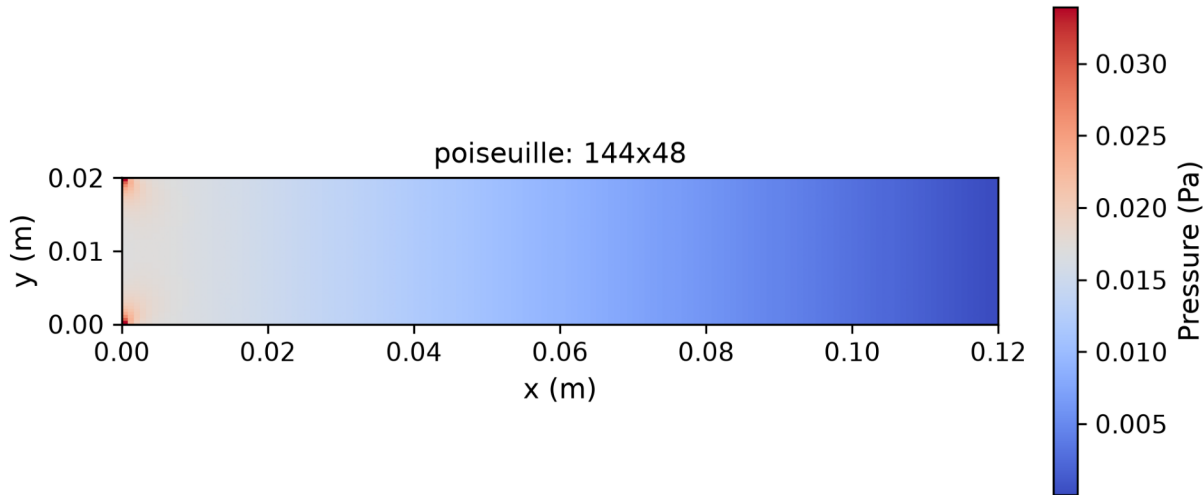
LBM 32x32, Ma=0.0249: u L2=0.64003%, p L2=1.68311%, passed=True

FVM advantages supported here are the incompressible pressure constraint, authoritative face continuity, discrete energy/work balance and no lattice Mach requirement on its physical time step. Primary fields per physical plane occupy 4N2 doubles versus D2Q9 9N2 doubles; this is not peak memory.

Observed CPU timings include setup, diagnostics and recording, with different z replication and time steps. No FVM speed or peak-memory superiority is claimed. LBM local updates avoid a global pressure solve. This is one BGK baseline, not all TensorLBM capabilities.

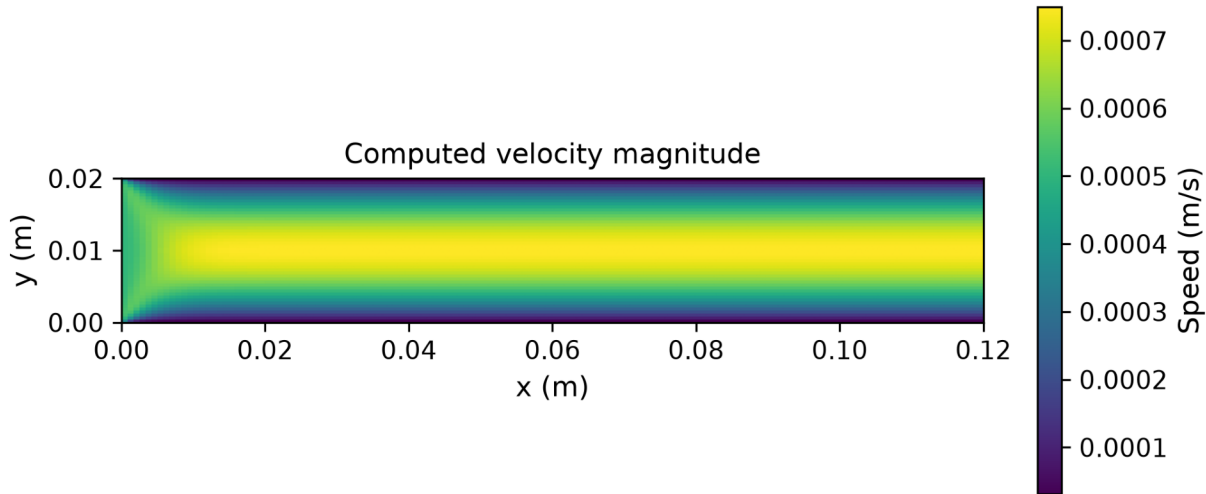
poiseuille, 144x48: pressure

Pressure and velocity have documented different clocks where applicable.



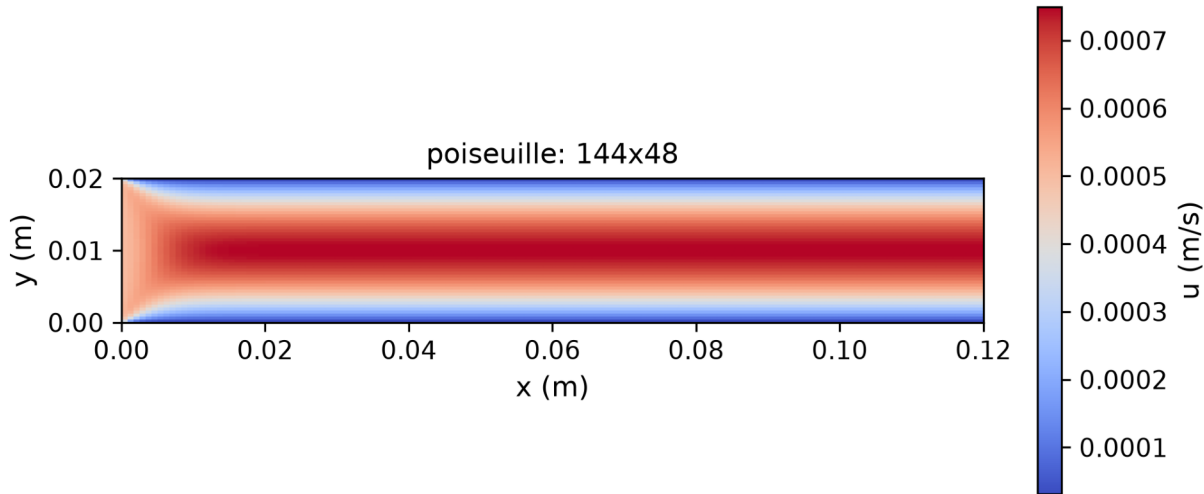
poiseuille, 144x48: speed

Pressure and velocity have documented different clocks where applicable.



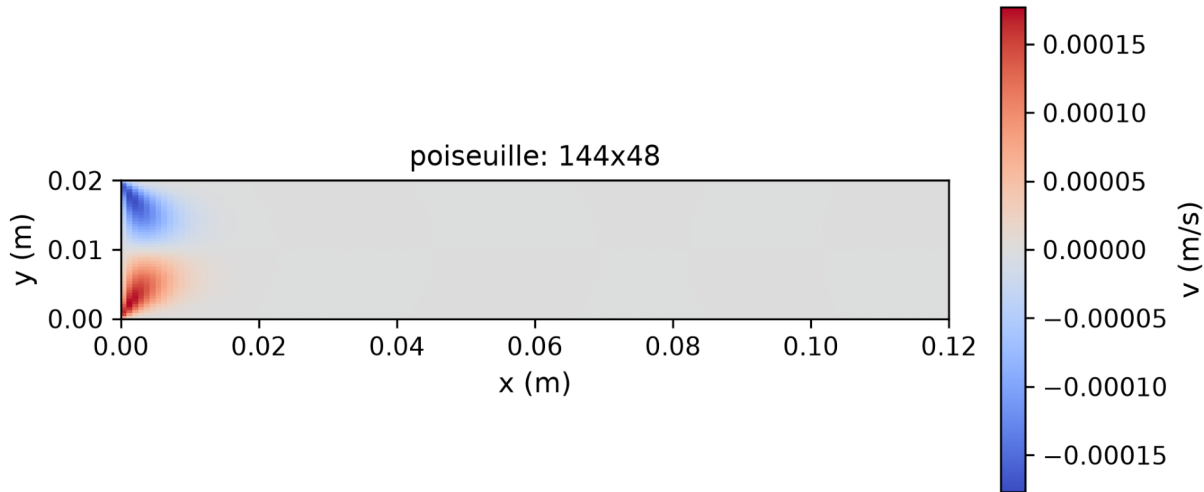
poiseuille, 144x48: u

Pressure and velocity have documented different clocks where applicable.



poiseuille, 144x48: v

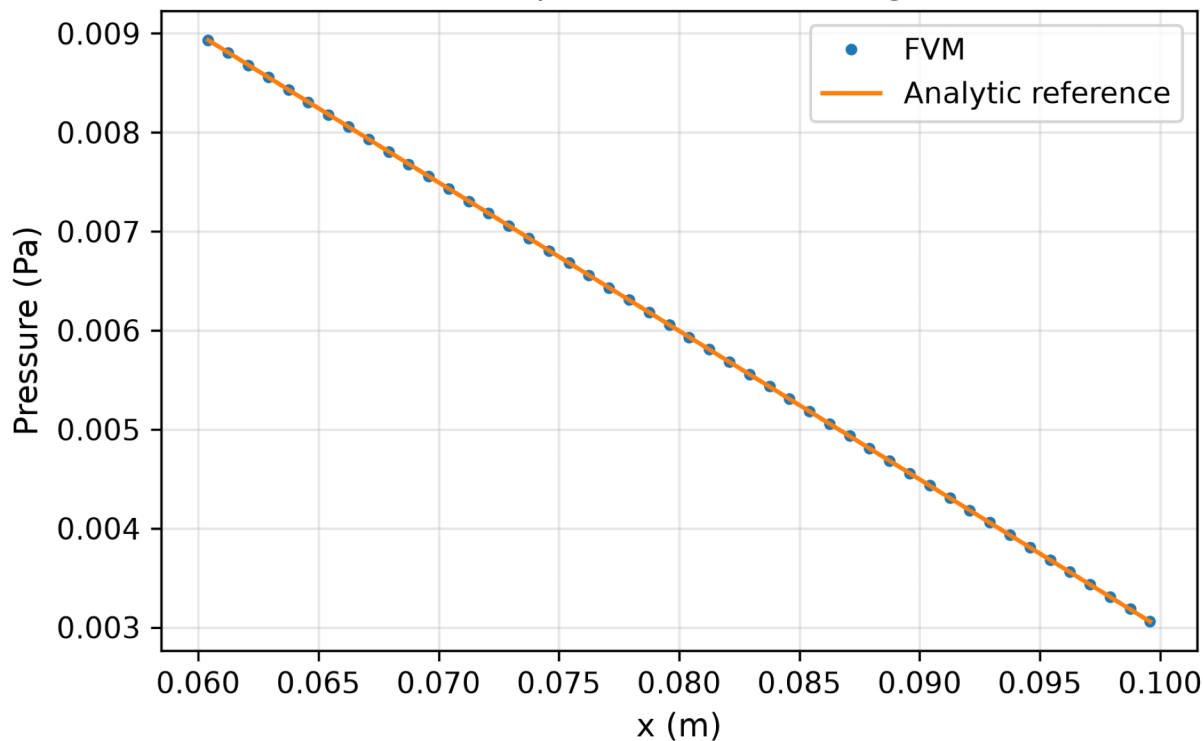
Pressure and velocity have documented different clocks where applicable.



poiseuille, 144x48: pressure-curve

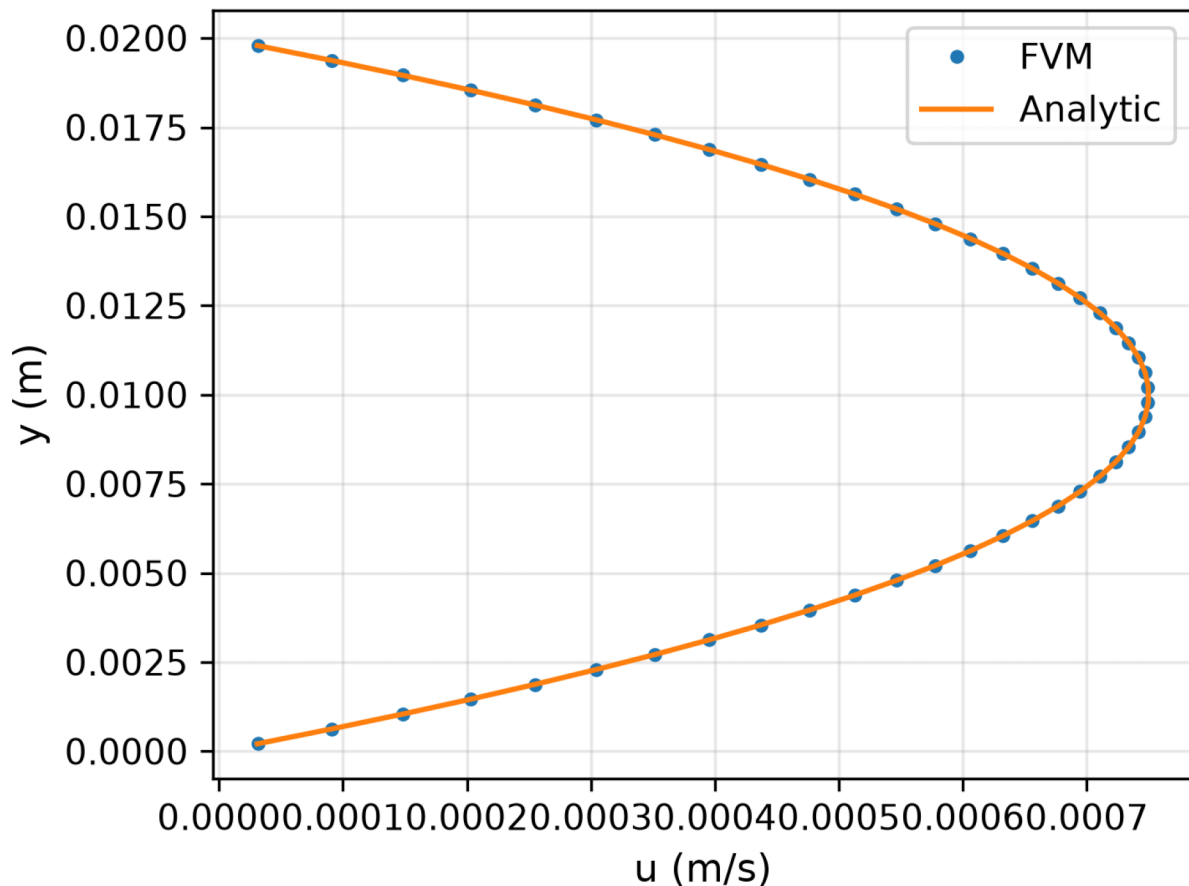
Pressure and velocity have documented different clocks where applicable.

Pressure comparison in declared region/time



Pressure and velocity have documented different clocks where applicable.

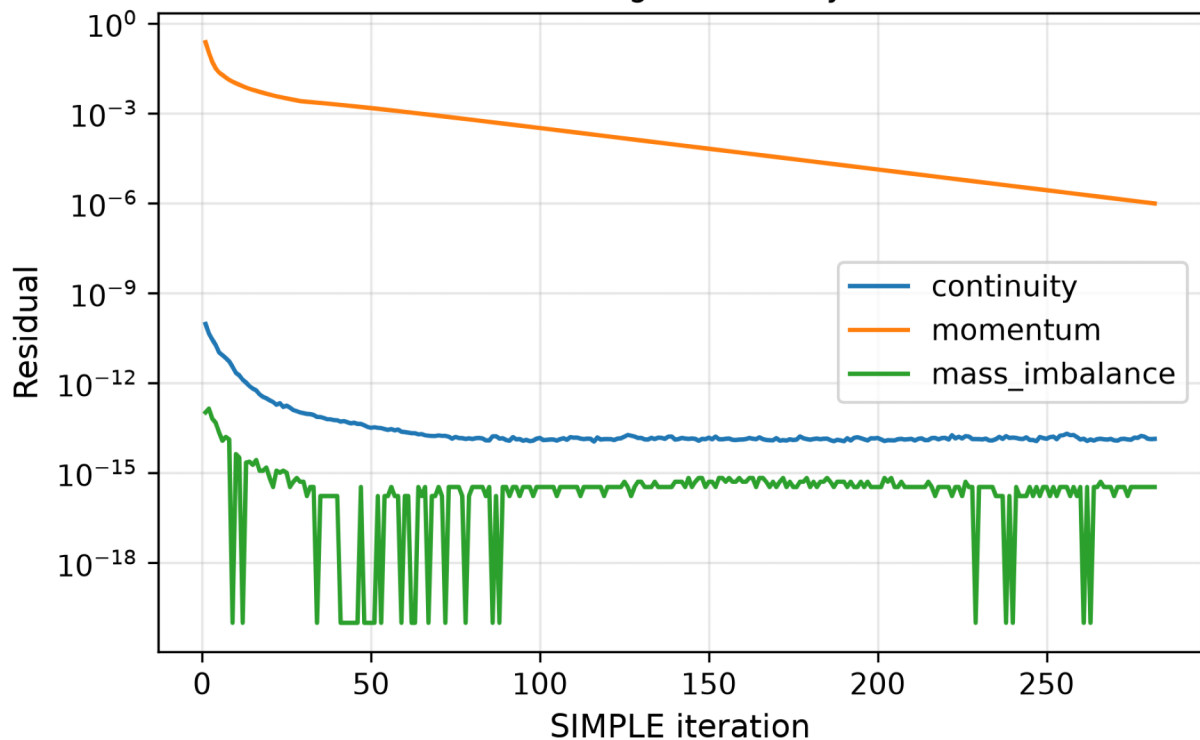
Velocity profile at documented sample location



poiseuille, 144x48: history

Pressure and velocity have documented different clocks where applicable.

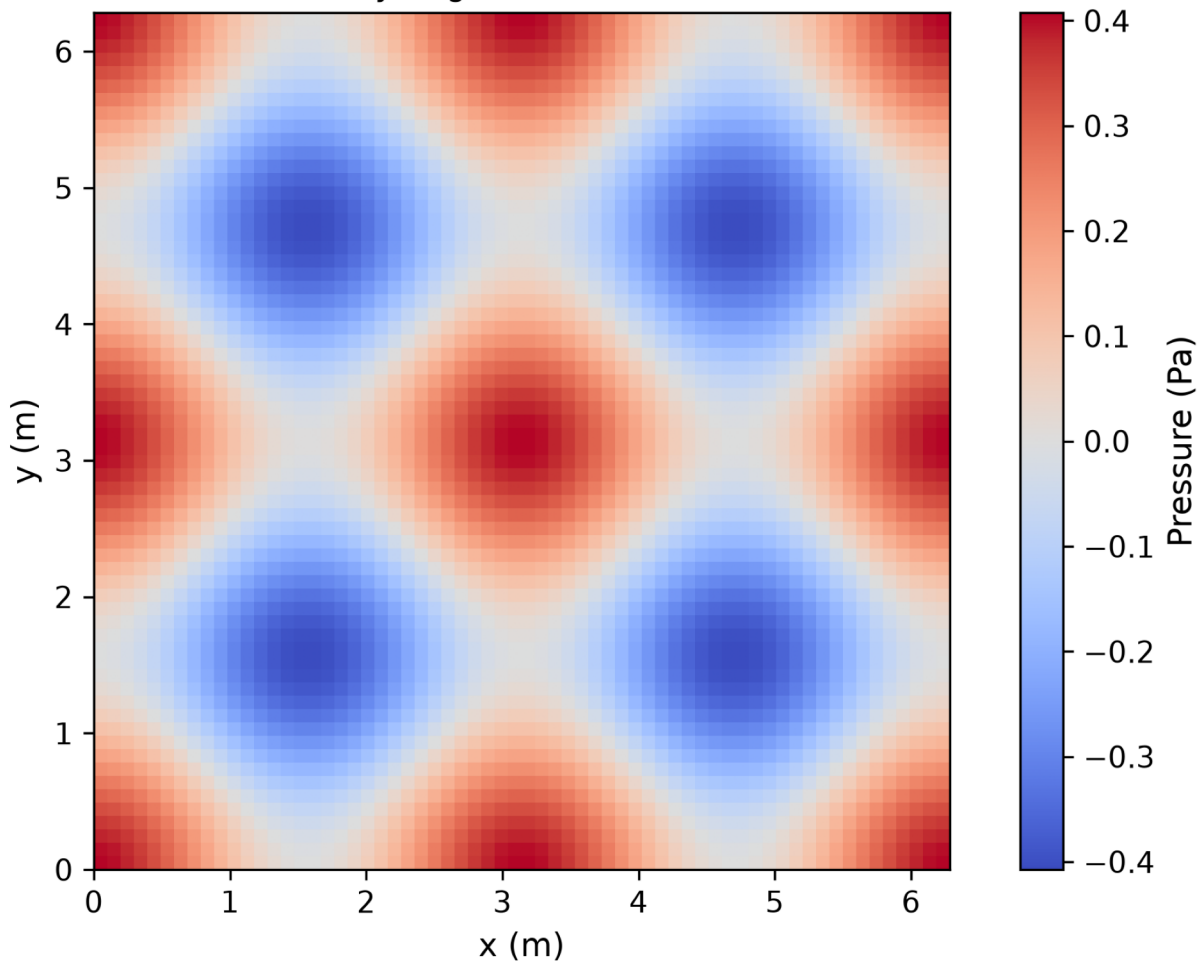
Convergence history



taylor-green, 64x64x4: pressure

Pressure and velocity have documented different clocks where applicable.

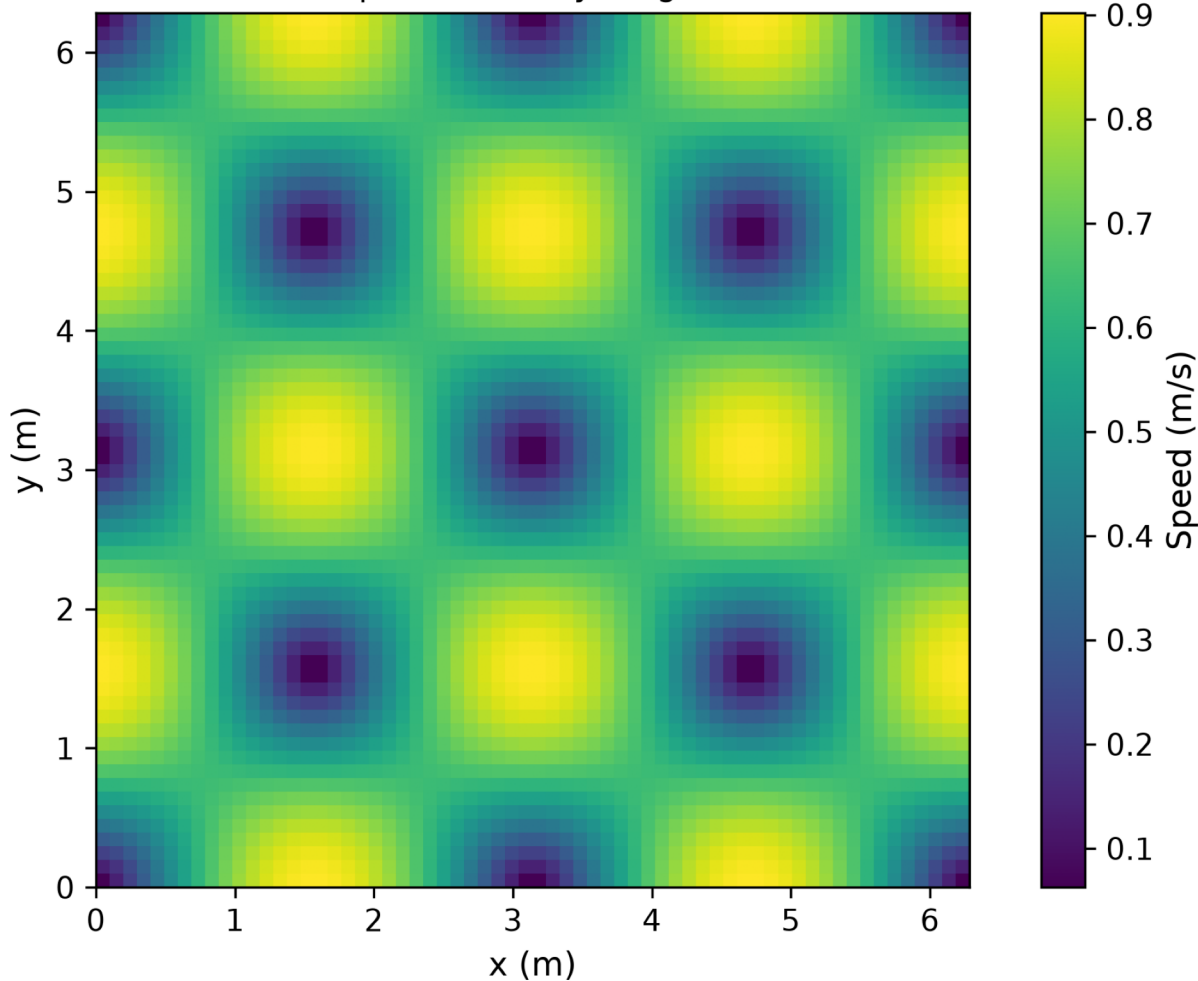
taylor-green: 64x64x4



taylor-green, 64x64x4: speed

Pressure and velocity have documented different clocks where applicable.

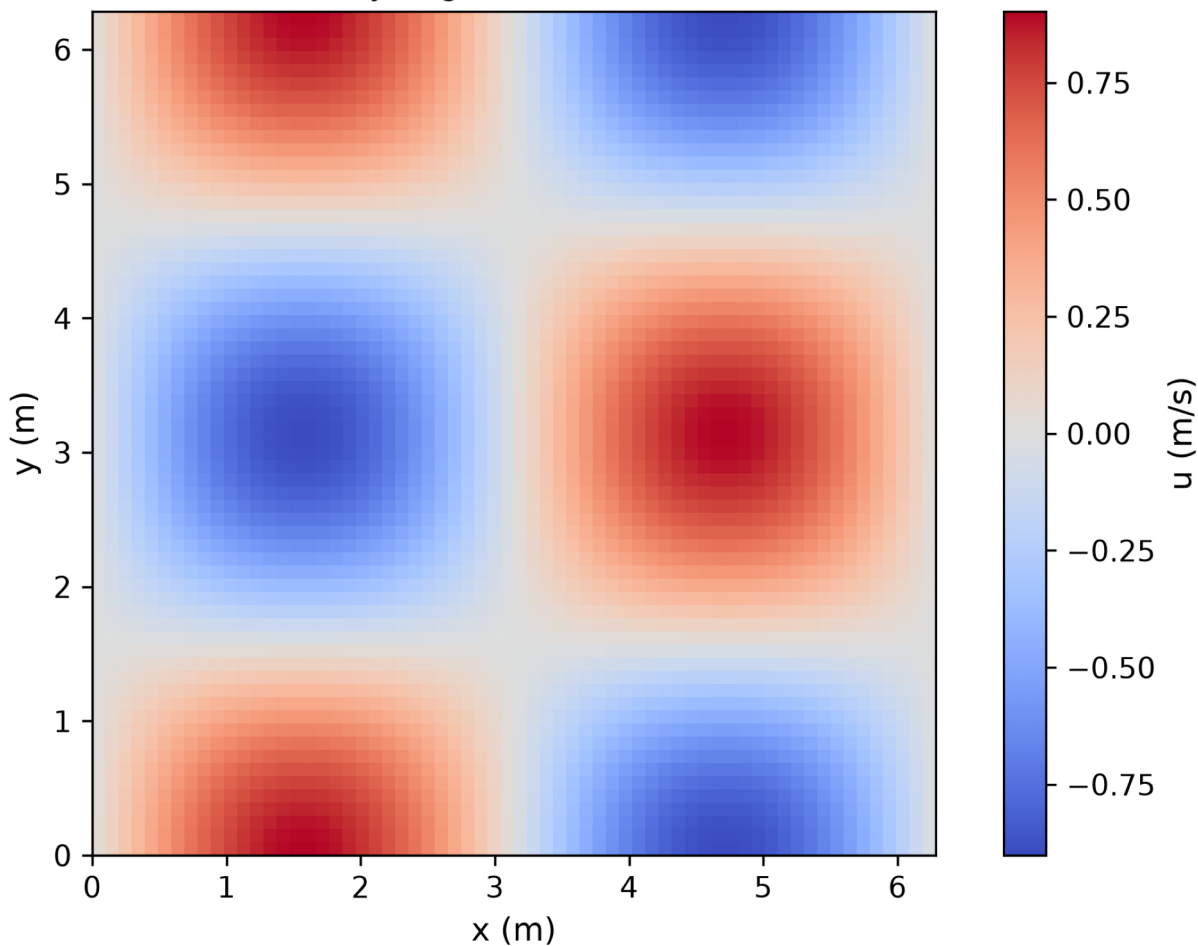
Computed velocity magnitude



taylor-green, 64x64x4: u

Pressure and velocity have documented different clocks where applicable.

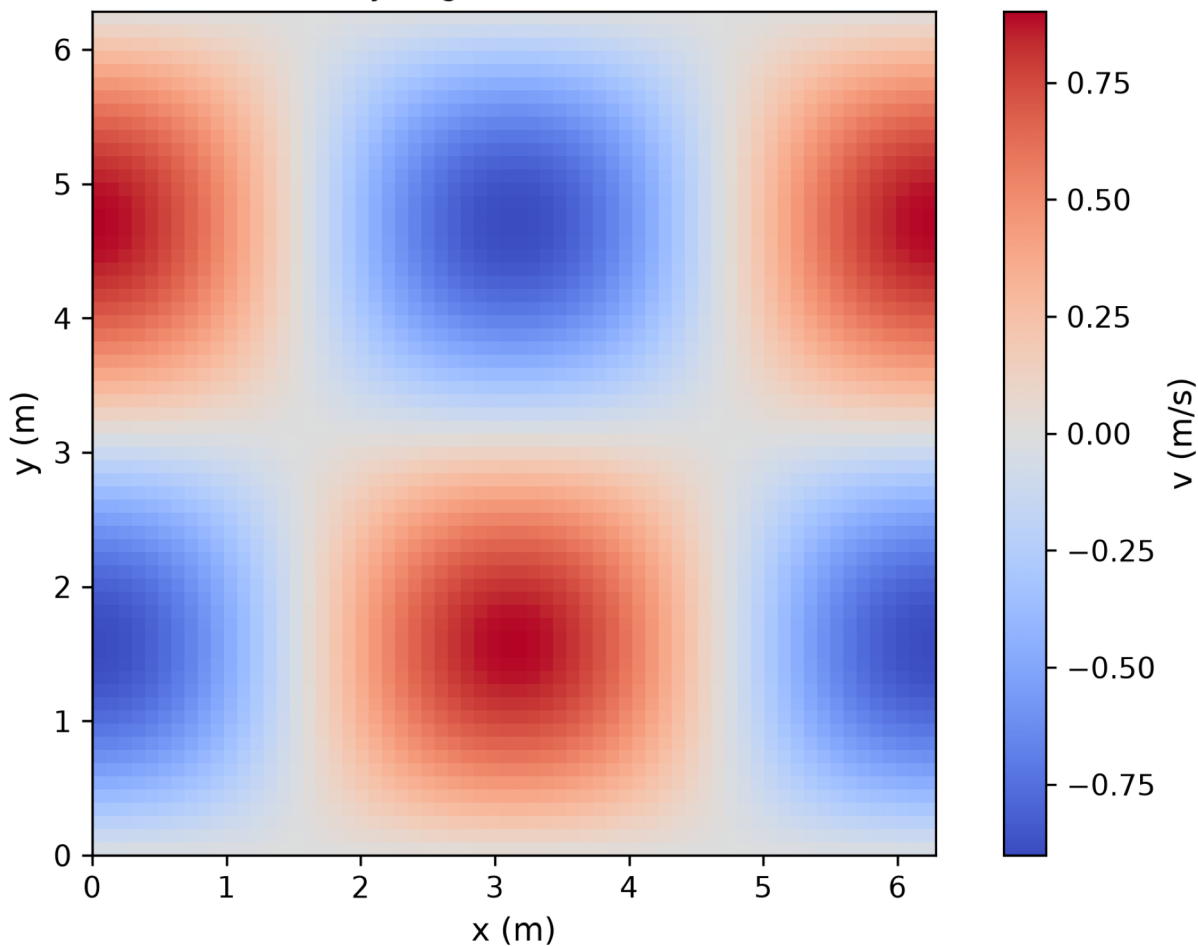
taylor-green: 64x64x4



taylor-green, 64x64x4: v

Pressure and velocity have documented different clocks where applicable.

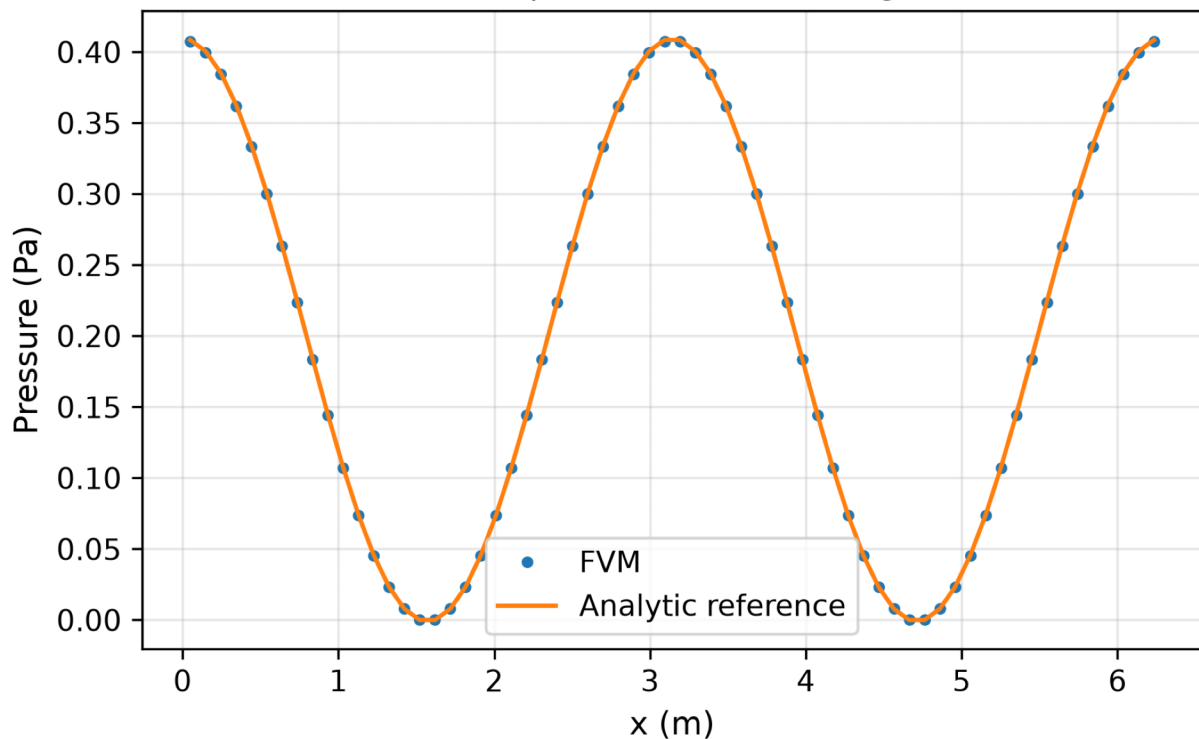
taylor-green: 64x64x4



taylor-green, 64x64x4: pressure-curve

Pressure and velocity have documented different clocks where applicable.

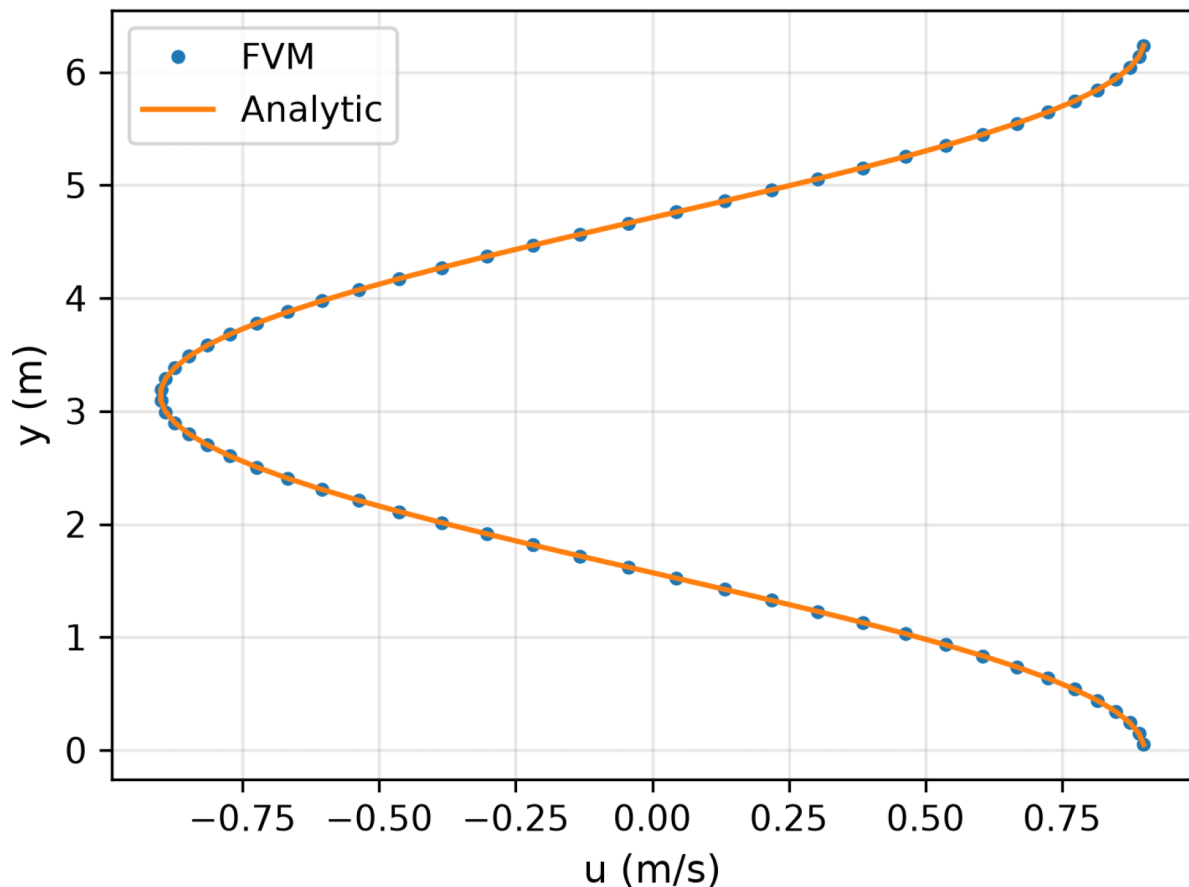
Pressure comparison in declared region/time



taylor-green, 64x64x4: velocity-profile

Pressure and velocity have documented different clocks where applicable.

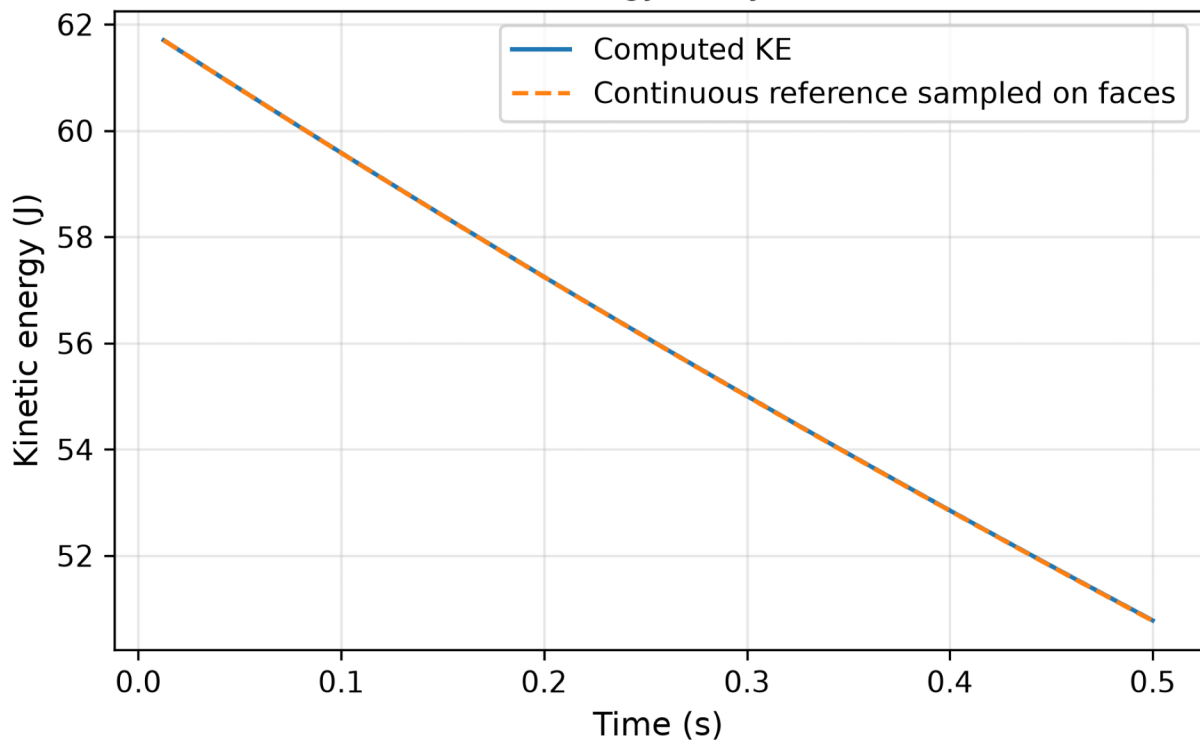
Velocity profile at documented sample location



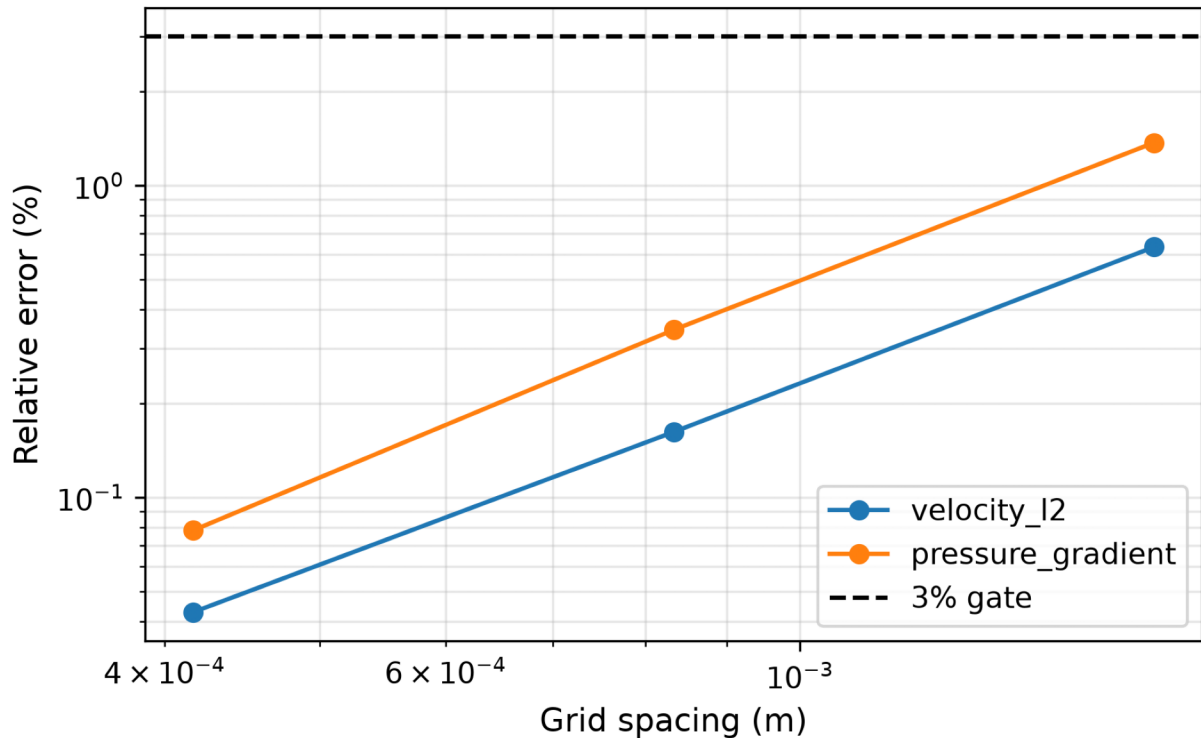
taylor-green, 64x64x4: history

Pressure and velocity have documented different clocks where applicable.

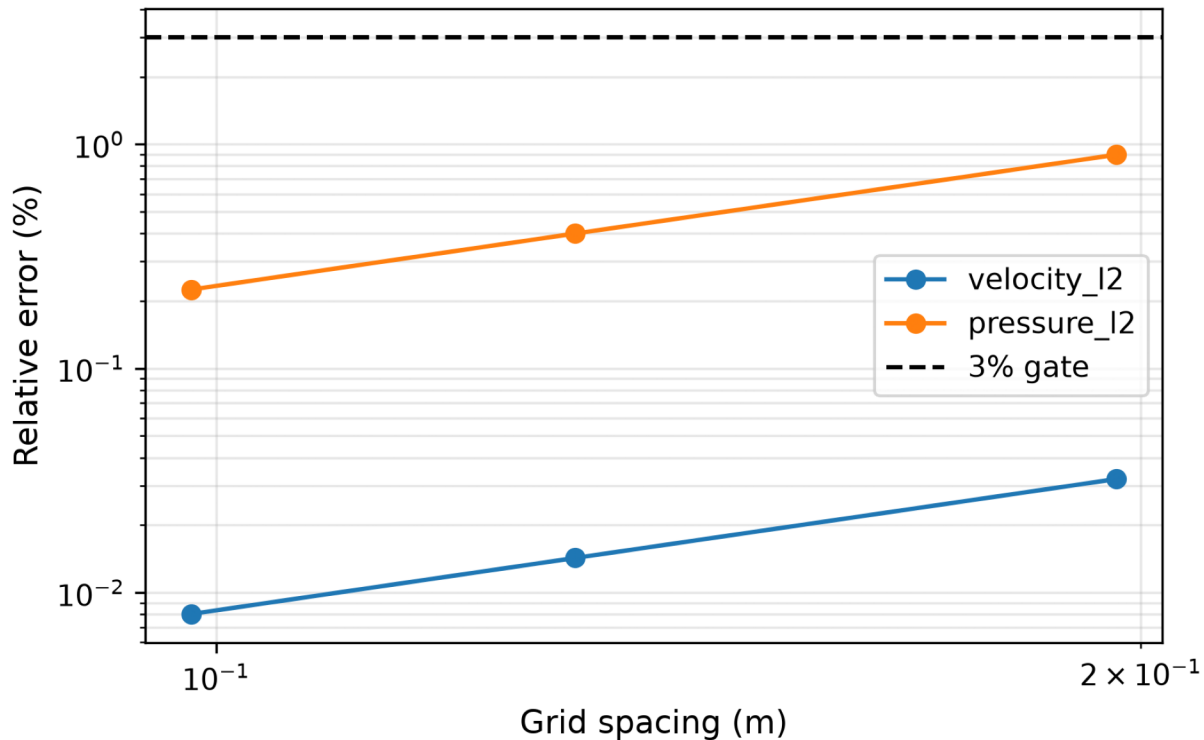
Energy decay



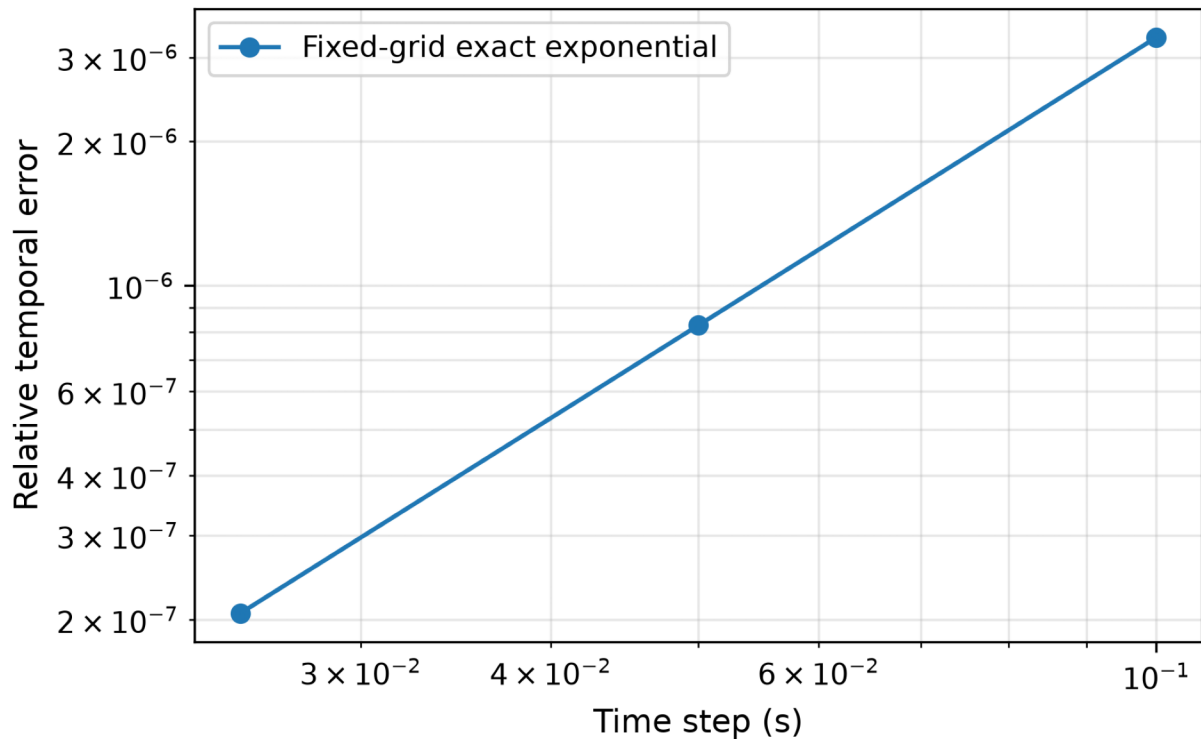
poiseuille: spatial refinement



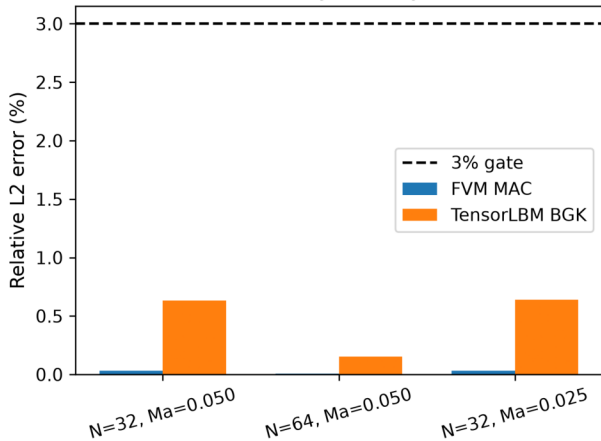
taylor-green: spatial refinement



Taylor-Green: temporal refinement



velocity accuracy



pressure accuracy

